

Aman Murari Singh

AI Researcher & Machine Learning Engineer

amanmuraris@gmail.com | +91-9229275147 | linkedin.com/in/amanmurarisingh | github.com/amanmurari

EDUCATION

B.Tech. in Computer Science and Engineering, Siksha 'O' Anusandhan University | 2024 - 2028 | CGPA: 8.71

EXPERIENCE

AI Engineer | Doubt Buddy

September 2025 - February 2026

- Engineered Datagentz, an AI-powered Microsoft Excel add-in that transforms raw spreadsheets into insights using natural language queries, automated pattern detection, and smart visualizations.
- Built AI workflows for spreadsheet understanding, PDF-to-data extraction, data transformation, conversational analytics, and Excel-based automation.
- Integrated AI capabilities into Excel's native environment, enabling non-technical users to perform advanced data analysis without complex formulas.
- Worked on end-to-end AI product development, including feature design, backend integration, workflow optimization, and Microsoft Marketplace publication.

AI Research Intern | Vizuara Technologies Private Limited

May 2025 - October 2025

- Worked on nano-GPT-OSS, an open-source research project focused on efficient Transformer architecture design and small-scale language model training.
- Implemented hybrid attention mechanisms combining sliding-window attention and sink tokens to improve model efficiency and context handling.
- Integrated MoE-enhanced MLP blocks and achieved 12-30% lower validation loss compared to GPT-2 across 6-12 layer model configurations.
- Reduced RAM usage by 11% and disk footprint by 11%; demonstrated 30.5% validation loss improvement at 12 layers.
- Implemented PyTorch Distributed Data Parallel for multi-GPU language model pretraining and open-sourced code, logs, and metrics, helping the repository reach 100+ GitHub stars and 30+ forks.

PROJECTS

nanoKimi: Lightweight LLM Research Pipeline

- Built a compact LLM research pipeline inspired by nanoGPT and Kimi-K2 for efficient training and inference experimentation.
- Integrated Muon optimizer, Mixture of Experts, and Latent Attention; benchmarked training speed, inference throughput, model quality, and parameter efficiency. GitHub: github.com/amanmurari/nanokimi

Multimodal Retrieval-Augmented Generation System

- Built a multimodal RAG system for text and image retrieval using FAISS-based similarity search and embedding-based retrieval.
- Designed retrieval pipelines for document search, image search, and AI-powered question answering across multiple data formats.

Reinforcement Learning Autonomous Traffic Control System

- Developed a reinforcement learning-based traffic-control environment where agents learn signal-control strategies through feedback.
- Designed reward functions, traffic scenarios, state-action logic, and policy improvement flow to optimize traffic movement.

SKILLS

Programming: Python, JavaScript, HTML, CSS

AI/ML: Machine Learning, Deep Learning, Reinforcement Learning, NLP, Computer Vision, Generative AI

LLMs: Transformers, RAG, Fine-tuning, Multi-Agent Systems, Prompt Engineering, MoE, Attention Mechanisms

Frameworks and Tools: PyTorch, TensorFlow, FastAPI, Django, OpenCV, LangChain, CrewAI, AutoGen, FAISS, Docker, Git, GitHub, PostgreSQL, MySQL, Pandas, NumPy, Scikit-learn, XGBoost

ACHIEVEMENTS

- Won 1st place at AI/ML Hackathon Synchronize 4.0, competing against 100+ teams.
- Won 1st place at the Vizuara AI Research Challenge, a national-level hackathon with 100+ teams, earning a two-month paid AI Research Internship.
- Selected as a finalist at UIDAI Data Hackathon 2026, a national-level hackathon with 1,000+ teams.
- Solved 150+ problems on LeetCode; core contributor to nano-GPT-OSS, an open-source LLM research project with 100+ GitHub stars and 30+ forks.

CERTIFICATIONS

TensorFlow Bootcamp | Data Structures and Algorithms in Java - Apna College | Multi-Agent AI Systems Certification

LANGUAGES

English, Hindi